

PERIODIC SIGNINGS OF $C_N(1, 2)$: AN EXACT BAND EDGE AND SHORT-PERIOD CLASSIFICATION

Lennart Rudolph 

ABSTRACT. We study spectral-radius optimization for periodic real signings of the 4-regular circulants $C_n(1, 2)$. After gauging all step-1 edges positive, a period-8 triangle-flux pattern produces an explicit 8×8 Hermitian Bloch symbol. Its upper band edge occurs at phase $z = 1$, and the associated infinite operator and every finite quotient $C_{8q}(1, 2)$ have the same spectral radius, where t_0 is the largest root displayed below:

$$\sqrt{t_0} = 2.7936044933 \dots, \quad t_0^4 - 16t_0^3 + 76t_0^2 - 96t_0 + 16 = 0.$$

The proof uses an exact Bloch determinant, rational Sturm certificates, and monotonicity on the full unit-circle phase interval. Independently, a symmetry-reduced certificate classifies all triangle-flux patterns of fundamental period at most 16: the period-8 pattern is the unique norm minimizer up to rotation, reflection, and global flux complement. As an application, its radius is strictly below Suvagiya's proposed twisted-class value for every $n = 8q \geq 32$, disproving the corresponding signed-circulant minimum conjecture on an infinite arithmetic progression. All decisive comparisons and classification witnesses are exact; floating-point linear algebra is used only for discovery and stress testing.

1. INTRODUCTION

For a graph G and a signing $\sigma: E(G) \rightarrow \{\pm 1\}$, let A_σ be the signed adjacency matrix. Switching and balance are classical organizing notions for signed graphs [7], and finding a signing with small spectral radius is a basic problem in spectral graph theory; see [1] for broader context. Through the 2-lift correspondence, the eigenvalues of a signed adjacency matrix control the new eigenvalues of a graph lift. This viewpoint underlies the program of Bilu and Linial [2] and the bipartite Ramanujan construction of Marcus, Spielman, and Srivastava [4]. The underlying optimization problem is

$$\min_{\sigma \in \{\pm 1\}^{E(G)}} \rho(A_\sigma), \quad \rho(A) = \max\{|\lambda| : \lambda \in \text{spec}(A)\}.$$

Switching at vertices preserves the spectrum, but even after passing to switching classes the search space is exponential.

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Periodic signings add useful structure to this optimization problem. A gauge can move the signs of a one-dimensional periodic graph into a short flux word, and Floquet–Bloch theory replaces the infinite operator by a finite Hermitian matrix symbol over the unit circle [3]. Finite cyclic quotients then sample that same symbol at roots of unity. This framework makes it possible to prove a continuum band-edge statement and every finite-quotient statement from one exact determinant.

We carry this program out for the 4-regular circulant

$$C_n(1, 2) = (\mathbb{Z}/n\mathbb{Z}, \{\{i, i+1\}, \{i, i+2\} : i \in \mathbb{Z}/n\mathbb{Z}\}).$$

The main construction is a period-8 triangle-flux word. We compute its full 8×8 Bloch determinant, prove that the spectral edge over all phases is attained at $z = 1$, and identify the common norm of the infinite operator and all finite quotients $C_{8q}(1, 2)$. We then classify, by an exact symmetry-reduced certificate, every flux word of fundamental period at most 16. The classification shows that the construction is not an isolated search hit: it is the unique short-period norm minimizer up to the natural dihedral and complement symmetries.

This analysis has an immediate extremal application. Suvagiya [6] studies the same signing problem on $C_n(1, 2)$. For even n , a distinguished twisted switching class has spectral radius

$$\rho_-(n) = 2\sqrt{\cos^2(\pi/n) + \cos^2(2\pi/n)}.$$

Exhaustive computations for $n \in \{8, 10, 12, 14, 16, 18\}$ led to the conjecture that this value is the minimum over all signings for every even $n \geq 8$ [6, Conjecture 3]. The construction below disproves that statement at $n = 32$ and on an infinite arithmetic progression.

A companion parity-family study reports restricted-family experiments at larger orders [5]. Its computational-evidence section reports 2.790 on $C_{30}(1, 2)$ after solving the even-cycle constraints and sampling the resulting affine family, compared with 2.856 from best-of-60 random draws followed by local search. The first value agrees, to the quoted precision, with $\rho_-(30) = 2.7898720610\dots$; neither experiment is reported there as a global enumeration over all switching classes.

All proof-critical computations have small exact certificates. One standard-library verifier recomputes the characteristic polynomial at $n = 32$, the Sturm counts, and the Bloch determinant. A separate verifier checks the short-period classification using only integer arithmetic. Floating-point linear algebra is used to find candidate witnesses, never to accept one.

2. A PERIODIC SIGNING AND ITS FINITE CONSEQUENCE

Index vertices by $\mathbb{Z}/n\mathbb{Z}$. Give every step-1 edge $\{i, i+1\}$ sign $+1$, and give the step-2 edge $\{i, i+2\}$ sign β_i , where

$$(1) \quad (\beta_0, \dots, \beta_7) = (+1, -1, +1, -1, -1, +1, -1, +1)$$

is repeated periodically. Equivalently, $\beta_i = -1$ precisely when $i \bmod 8 \in \{1, 3, 4, 6\}$. Let H_β be the corresponding bounded self-adjoint operator on $\ell^2(\mathbb{Z})$, and let A_{8q} be its quotient on $C_{8q}(1, 2)$.

Theorem 2.1. *For every integer $q \geq 1$,*

$$\|H_\beta\| = \rho(A_{8q}) = \sqrt{t_0} = 2.7936044933348411065\dots,$$

where t_0 is the largest root of

$$(2) \quad Q(t) = t^4 - 16t^3 + 76t^2 - 96t + 16.$$

For every $q \geq 4$,

$$\rho(A_{8q}) < \rho_-(8q).$$

Consequently Conjecture 3 of [6] is false for every $n \equiv 0 \pmod{8}$ with $n \geq 32$.

The exact finite certificate for the base quotient $n = 32$ is already concise.

Proposition 2.2. *For $n = 32$, the characteristic polynomial of A_{32} is*

$$\begin{aligned} \chi_{A_{32}}(x) = & (x^4 - 12x^2 + 34)(x^4 - 4x^2 + 2) \\ & \cdot (x^4 - 2x^3 - 6x^2 + 12x - 4) \\ & \cdot (x^4 + 2x^3 - 6x^2 - 12x - 4) \\ & \cdot (x^8 - 16x^6 + 80x^4 - 128x^2 + 38)^2. \end{aligned}$$

Every root lies strictly between $-1397/500$ and $1397/500$, while $1397/500 < \rho_-(32)$.

Proof. Integer Faddeev–LeVerrier computation gives the displayed factorization. For each of the four quartic factors, the Sturm variation counts at $-1397/500$ and $1397/500$ are 4 and 0; for the octic factor they are 8 and 0. Hence all roots of every factor lie in the claimed interval.

For the other inequality, elementary squaring gives

$$\sqrt{2} > 1.414, \quad \sqrt{2 + \sqrt{2}} > 1.847, \quad \sqrt{2 + \sqrt{2 + \sqrt{2}}} > 1.961.$$

The half-angle identities therefore imply

$$\begin{aligned} \rho_-(32)^2 &= 4 + \sqrt{2 + \sqrt{2}} + \sqrt{2 + \sqrt{2 + \sqrt{2}}} \\ &> 4 + 1.847 + 1.961 = 7.808, \end{aligned}$$

whereas

$$(1397/500)^2 = 7.806436 < 7.808.$$

Every comparison is over the integers after clearing denominators. $\square$

The quartic $x^4 - 2x^3 - 6x^2 + 12x - 4$ is irreducible over $\mathbb{Q}$: its reduction $x^4 + x^3 + 2$ is irreducible over $\mathbb{F}_3$. Thus it is the minimal polynomial of $\sqrt{t_0}$, with the spectral radius distinguished as its largest real root.

3. EXACT BLOCH BAND EDGE AND FINITE SAMPLING

Let $z \in \mathbb{T} = \{w \in \mathbb{C} : |w| = 1\}$ and let $B(z)$ be the 8×8 Hermitian Bloch matrix of the period-8 operator defined by (1). Explicitly, if a_{ij} denotes the signed edge weight on the infinite graph and $0 \leq r, s < 8$, then

$$B(z)_{rs} = \sum_{m \in \mathbb{Z}} a_{r,s+8m} z^m.$$

Only finitely many summands are nonzero, and $B(z)$ is Hermitian for $|z| = 1$. With rows and columns ordered $0, 1, \dots, 7$, the matrix is

$$B(z) = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & -z^{-1} & z^{-1} \\ 1 & 0 & 1 & -1 & 0 & 0 & 0 & z^{-1} \\ 1 & 1 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & -1 & 1 & 0 & 1 & 1 \\ -z & 0 & 0 & 0 & -1 & 1 & 0 & 1 \\ z & z & 0 & 0 & 0 & 1 & 1 & 0 \end{pmatrix}.$$

Writing μ for normalized Haar measure on $\mathbb{T}$, the cellwise Fourier transform gives the two precise spectral reductions used below:

$$(3) \quad H_\beta \cong \int_{\mathbb{T}}^\oplus B(z) d\mu(z), \quad A_{8q} \cong \bigoplus_{z^q=1} B(z).$$

Consequently $\|H_\beta\| = \max_{z \in \mathbb{T}} \rho(B(z))$ and $\rho(A_{8q}) = \max_{z^q=1} \rho(B(z))$. Thus the determinant calculation below can be reproduced directly from the displayed symbol. Expansion over $\mathbb{Z}[x, z, z^{-1}]$ gives

$$(4) \quad \det(xI - B(z)) = F(x^2, z + z^{-1}),$$

where

$$(5) \quad F(t, y) = t^4 - 16t^3 + (80 - 2y)t^2 + (-128 + 16y)t + y^2 - 13y + 38.$$

At $z = 1$, hence $y = 2$, this specializes to $F(t, 2) = Q(t)$.

We isolate the largest root of Q without numerical assumptions. Set

$$L = \frac{39}{5} = 7.8, \quad H = \frac{1561}{200} = 7.805.$$

Then

$$Q(L) = -\frac{179}{625}, \quad Q(H) = \frac{84332641}{1600000000}.$$

A rational Sturm sequence gives one root in (L, H) and no root in $[H, \infty)$; it also confirms that all four roots of Q are real. Thus $L < t_0 < H$.

For $t \geq L$ and $-2 \leq y \leq 2$,

$$(6) \quad \frac{\partial F}{\partial y} = 2y - 2t^2 + 16t - 13 < 0,$$

$$(7) \quad \frac{\partial F}{\partial t} \geq \frac{\partial F}{\partial t}(t, 2) = 4(t - 4)(t^2 - 8t + 6) > 0.$$

Indeed, the upper bound $-2t^2 + 16t - 9$ in (6) is already $-147/25$ at L and decreases thereafter. For (7), subtracting the value at $y = 2$ leaves $(16 - 4t)(y - 2) \geq 0$, and both displayed factors are positive at and above L .

It follows first that $F(t_0, y) \geq F(t_0, 2) = 0$, with strict inequality when $y < 2$, and then that $F(t, y) > 0$ whenever $t > t_0$ and $-2 \leq y \leq 2$. If x is an eigenvalue of $B(z)$, then $F(x^2, z + z^{-1}) = 0$; hence $x^2 \leq t_0$. At $z = 1$, $F(x^2, 2) = Q(x^2)$, so $x = \pm\sqrt{t_0}$ are eigenvalues. Therefore

$$\rho(B(z)) \leq \sqrt{t_0} \quad (z \in \mathbb{T}), \quad \rho(B(1)) = \sqrt{t_0},$$

and (3) gives $\|H_\beta\| = \sqrt{t_0}$. The finite equality now follows in both directions, rather than from phase inclusion alone:

$$\rho(A_{8q}) = \max_{z^q=1} \rho(B(z)) \leq \max_{z \in \mathbb{T}} \rho(B(z)) = \sqrt{t_0}, \quad \rho(A_{8q}) \geq \rho(B(1)) = \sqrt{t_0},$$

because $1^q = 1$. Thus every finite quotient has exactly the same spectral radius. Finally,

$$t_0 < H = 7.805 < 7.808 < \rho_-(32)^2,$$

and $\rho_-(n)$ is increasing for $n \geq 8$ because both angles lie in $[0, \pi/2]$ and decrease with n . This proves Theorem 2.1.

4. SUPPLEMENTARY EXACT OPTIMIZATION THROUGH PERIOD 16

The preceding sections already give a complete, human-checkable disproof. A separate computer-assisted question is whether (1) is an isolated witness or a short-period optimum. The answer is exact in the latter sense.

For a general periodic sequence $\beta_i \in \{\pm 1\}$, retain the notation H_β for the operator on $\ell^2(\mathbb{Z})$ with every step-1 edge positive and the step-2 edge beginning at i signed by β_i . Any periodic signing of the infinite graph can be put in this form by switching along the step-1 path. If the original step-1 and step-2 signs are a_i and b_i , respectively, then the resulting $\beta_i = b_i a_i a_{i+1}$ is the switching-invariant sign of the triangle on $i, i+1, i+2$ and is periodic with the same declared period.

Rotation and reflection of β give unitarily equivalent operators. Also $\|H_\beta\| = \|H_{-\beta}\|$: conjugation by the diagonal operator $e_i \mapsto (-1)^i e_i$ negates the step-1 terms, after which multiplying the whole operator by -1 changes β to $-\beta$ without changing its norm. These are the natural flux symmetries used below.

Theorem 4.1. *Among all signings whose triangle-flux sequence has fundamental period at most 16, the pattern (1) uniquely minimizes the norm, up to rotation, reflection, and the global flux complement $\beta \mapsto -\beta$. The minimum is $\sqrt{t_0}$. More sharply, every other inequivalent pattern satisfies*

$$\|H_\beta\| > \frac{14}{5},$$

whereas the pattern (1) satisfies $\|H_\beta\| < 1397/500 < 14/5$.

Proof. The exceptional upper bound follows from the preceding section. For every period $p \leq 16$, encode β as a p -bit word and select the least word in its orbit under rotation, reflection, and complement. The exhaustive list contains 2626 canonical representatives across the sixteen period enumerations; this count deliberately includes words represented again at a nonfundamental multiple of their period. Exactly two entries reduce to the period-8 exceptional orbit, at declared periods 8 and 16.

For each of the other 2624 representatives, the certificate supplies a phase $z \in \{1, i, -1, -i\}$ and a nonzero Gaussian-integer vector $v \in \mathbb{Z}[i]^p$. Let $B_p(z)$ be the corresponding $p \times p$ Hermitian Bloch matrix. The checker reconstructs $B_p(z)$ and verifies, by exact integer arithmetic, that

$$(8) \quad 5|v^* B_p(z) v| > 14 v^* v.$$

The Rayleigh principle and the Bloch decomposition imply $\|H_\beta\| > 14/5$. For the supplied certificate, the smallest normalized strict excess in (8) among all witnesses is $202/49210$, so no equality or rounding case is hidden. Regenerating discovery vectors may produce a different positive margin without changing the exact statement. The verifier also regenerates the canonical orbit set from all 2^p words for every $p \leq 16$ and checks that the certificate covers it exactly once. This proves both completeness and the claimed uniqueness. $\square$

The enumeration uses $O(p2^p)$ orbit operations at period p when cyclic words are represented by machine integers. Numerical eigensolvers were used only to discover short vectors for (8); after a vector is rounded to $\mathbb{Z}[i]^p$, its acceptance, the orbit reduction, and the coverage check are entirely exact. This separation gives a compact workflow for other periodic signing searches: approximate discovery followed by a proof object whose verifier has no numerical-linear-algebra dependency.

5. DISCUSSION, REPRODUCIBILITY, AND SCOPE

The periodic construction has two roles independent of the motivating conjecture. First, it gives an exact band-edge example in which one Laurent determinant controls an infinite operator and every cyclic quotient. Second, Theorem 4.1 converts a numerical pattern search into a complete short-period optimization theorem with compact proof objects. The same workflow can be applied to other periodic signed graphs: gauge to flux data, reduce by geometric symmetries, locate candidate bands numerically, and retain only exact determinant, Sturm, or Rayleigh certificates.

The natural remaining questions are structural. The present classification does not decide whether a longer-period word has smaller norm, nor whether the period-8 word minimizes the signing problem on any fixed finite quotient. It does isolate the first unresolved search boundary: fundamental periods greater than 16, rather than another representation of a shorter word.

The accompanying artifact contains two independent exact verifiers. The first uses only the Python standard library and recomputes the 32×32 characteristic polynomial by Faddeev–LeVerrier, implements rational Sturm sequences, and expands the Laurent-polynomial determinant in (4). A separate SymPy program provides a cross-check. The supplied classification certificate, identified by SHA-256 in the artifact manifest, is checked by a second standard-library program that reconstructs every Gaussian-integer Rayleigh quotient and independently enumerates all symmetry orbits. Direct numerical matrices through $n = 256$, Bloch blocks through 251 cells, and randomly sampled off-grid phases provide additional stress tests but are not used as proof.

A standalone Lean project, pinned to Lean 4.27.0 and an exact Mathlib commit recorded in its resolved manifest, formalizes the rational separators, radical lower bounds, derivative inequalities (6)–(7), and one concrete Bloch-algebra bridge. At phase $z = 1$, it checks that the displayed 8×8 block exchanges two explicit rank-4 lattices, that its square reduces to a specified 4×4 integer block, and that this block has characteristic polynomial Q . Its scope is deliberately limited: the exact executables check the 32×32 characteristic polynomial, Sturm root counts, and full phase-dependent Bloch determinant, while the Fourier reductions and spectral-theorem steps are manuscript arguments. None of these remaining steps is claimed as fully formalized in Lean.

Theorem 4.1 determines the optimum only among flux patterns of fundamental period at most 16. It does not assert global optimality among all periods, nor does it identify the minimum over every signing of a fixed $C_n(1, 2)$. These remain separate optimization questions. No prior resolution of Conjecture 3 or substantially equivalent periodic construction was found in the documented literature and repository sweep through August 11, 2026. As with any finite search, this is not an absolute priority claim.

DATA AND CODE AVAILABILITY

The exact verification artifact described in Section 5 is supplied with this manuscript as supplementary material. It contains the source code, the period-at-most-16 classification certificate, pinned dependencies, SHA-256 hashes, and reproduction instructions. The versioned archival record containing the preprint, source, and exact verification artifact is available at <https://doi.org/10.5281/zenodo.21892995>.

DECLARATION OF AGENTIC AI AND AI-ASSISTED TECHNOLOGIES

During the discovery, verification, and preparation of this work, the author used OpenAI Codex and Anthropic Claude as agentic research assistants. Codex assisted with computational exploration, literature and novelty searches, code development and testing, proof development, and manuscript drafting and editing. Claude was used for independent adversarial review,

including attempts to identify mathematical, computational, citation, and presentation defects. The author independently checked the claims, proofs, computations, sources, and final text, accepts full responsibility for the work, and confirms that neither system is an author.

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